

Historically biomes are known to move as climate changes. A classic example is the Sahara Desert, which years ago was supposed to be a lush landscape with river flowing through it. Accordingly, appropriate fauna like Hippos, Giraffes, Crocodiles lived amid abundant trees. Over course of time the climate dried out. It has now become the planets largest desert. The animals have migrated out to adjacent regions with more favourable conditions.

(Source: National Geography)

- Alpine zone of Himalayas is characterized by sparseness of animal groups. Many invertebrates of alpine zone are predatory and occur in lakes, streams and ponds. Among fishes, amphibians and vertebrates are totally lacking and reptilian fauna is greatly impoverished.
- Flora of alpines includes alpine phacelia, bear grass, bristlecone pine, moss campion, polylepis forest, pygmy bitterroot, and wild potato.

Forest biomes

Forest is a broad term used to describe areas where there are a large number of trees (**Fig. 10.8**). The forest biomes include a complex assemblage of different kinds of biotic communities. The major forest biomes are the Tropical forests and the Temperate forests.

Tropical forest

- They occur near the equator (between latitudes 23.5° at north and 23.5° at south).
- The major characteristic of tropical forests is their distinct seasons. Only two seasons are present (rainy and dry). Winter is absent. The length of daylight is about 12 hours and varies little.
- The average annual temperature ranges between 20° C and 25°C.
- Precipitation is evenly distributed throughout the year with annual rainfall exceeding 2000 mm.
- Soil is nutrient-poor and acidic. Decomposition is rapid and soils are subject to heavy leaching.
- Tree canopy is multilayered and continuous, allowing little light penetration.
- Flora is highly diverse: one square kilometer may contain as many as 100 different tree species. Trees are 25-35 m tall, with buttressed trunks and shallow roots, mostly evergreen, with large dark green leaves. Common vegetation are orchids, bromeliads, vines (lianas), ferns, mosses, and palms.
- They are characterized by the greatest diversity of fauna which includes birds, bats, small mammals, and insects.

More than half of earth's tropical forests have already been destroyed.



Fig. 10.8 Forest biome

Temperate forest

- These forests occur in eastern North America, northeastern Asia and western and central Europe.
- Have well-defined seasons with a distinct winter. Moderate climate and a growing season of 140-200 days during 4-6 frost-free months distinguish temperate forests.

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